

BẢN IN KỊCH BẢN HẠ TẦNG TERRAFORM IAAC:

main.tf

Hệ thống Quản lý và Số hóa Tài liệu Thư viện HCMUS (HCMUS-LDMS)

THÔNG TIN TẬP NGUỒN (FILE CONTROL)

Trường thông tin	Nội dung chi tiết
Tên tệp tin (File Name):	main.tf
Đường dẫn trong Repo (Path):	terraform/main.tf
Công nghệ:	HashiCorp Terraform (Infrastructure as Code - IaC)
Mô tả chức năng:	Kịch bản khởi tạo và cấu hình toàn bộ tài nguyên hạ tầng (Networks, Volumes, Containers)

1. Toàn văn mã nguồn kịch bản `terraform/main.tf`

```
# =====
# Terraform Main Configuration - HCMUS-LDMS Infrastructure as Code (IaaS)
# =====

terraform {
  required_version = "≥ 1.5.0"
  required_providers {
    docker = {
      source  = "kreuzwerker/docker"
      version = "~> 3.0"
    }
  }
}

provider "docker" {}

# --- Isolated Bridge Network ---

resource "docker_network" "ldms_network" {
  name      = "${var.project_name}_network"
  driver    = "bridge"
}

# --- Persistent Storage Volumes ---

resource "docker_volume" "postgres_data" {
  name = "${var.project_name}_postgres_data"
}

resource "docker_volume" "minio_data" {
  name = "${var.project_name}_minio_data"
}

resource "docker_volume" "grafana_data" {
  name = "${var.project_name}_grafana_data"
}

# --- Container Images ---

resource "docker_image" "postgres" {
  name          = "postgres:16"
  keep_locally = true
}

resource "docker_image" "minio" {
  name          = "minio/minio:latest"
  keep_locally = true
}

resource "docker_image" "mailhog" {
  name          = "mailhog/mailhog:latest"
  keep_locally = true
}
```

```

resource "docker_image" "prometheus" {
  name          = "prom/prometheus:latest"
  keep_locally = true
}

resource "docker_image" "grafana" {
  name          = "grafana/grafana:latest"
  keep_locally = true
}

# --- Infrastructure Services ---

# 1. PostgreSQL Database
resource "docker_container" "postgres" {
  name = "${var.project_name}_postgres"
  image = docker_image.postgres.image_id

  networks_advanced {
    name = docker_network.ldms_network.name
  }

  env = [
    "POSTGRES_USER=${var.postgres_user}",
    "POSTGRES_PASSWORD=${var.postgres_password}",
    "POSTGRES_DB=${var.postgres_db}"
  ]

  ports {
    internal = 5432
    external = var.postgres_port
  }

  volumes {
    volume_name      = docker_volume.postgres_data.name
    container_path = "/var/lib/postgresql/data"
  }

  restart = "unless-stopped"
}

# 2. MinIO S3 Object Storage
resource "docker_container" "minio" {
  name      = "${var.project_name}_minio"
  image     = docker_image.minio.image_id
  command = ["server", "/data", "--console-address", ":9001"]

  networks_advanced {
    name = docker_network.ldms_network.name
  }

  env = [
    "MINIO_ROOT_USER=${var.minio_root_user}",
    "MINIO_ROOT_PASSWORD=${var.minio_root_password}"
  ]

  ports {
    internal = 9000
  }

```

```

    external = var.minio_api_port
  }

  ports {
    internal = 9001
    external = var.minio_console_port
  }

  volumes {
    volume_name    = docker_volume.minio_data.name
    container_path = "/data"
  }

  restart = "unless-stopped"
}

# 3. MailHog SMTP Mock Service
resource "docker_container" "mailhog" {
  name = "${var.project_name}_mailhog"
  image = docker_image.mailhog.image_id

  networks_advanced {
    name = docker_network.ldms_network.name
  }

  ports {
    internal = 1025
    external = 1025
  }

  ports {
    internal = 8025
    external = var.mailhog_ui_port
  }

  restart = "unless-stopped"
}

# 4. Prometheus Metrics Monitoring
resource "docker_container" "prometheus" {
  name = "${var.project_name}_prometheus"
  image = docker_image.prometheus.image_id

  networks_advanced {
    name = docker_network.ldms_network.name
  }

  ports {
    internal = 9090
    external = var.prometheus_port
  }

  restart = "unless-stopped"
}

# 5. Grafana Monitoring Dashboard
resource "docker_container" "grafana" {

```

```
name = "${var.project_name}_grafana"
image = docker_image.grafana.image_id

networks_advanced {
  name = docker_network.ldms_network.name
}

env = [
  "GF_SECURITY_ADMIN_USER=admin",
  "GF_SECURITY_ADMIN_PASSWORD=admin",
  "GF_USERS_ALLOW_SIGN_UP=false"
]

ports {
  internal = 3000
  external = var.grafana_port
}

volumes {
  volume_name      = docker_volume.grafana_data.name
  container_path = "/var/lib/grafana"
}

restart = "unless-stopped"
}
```

BẢN IN KỊCH BẢN HẠ TẦNG TERRAFORM IAAC:

variables.tf

Hệ thống Quản lý và Số hóa Tài liệu Thư viện HCMUS (HCMUS-LDMS)

THÔNG TIN TẬP NGUỒN (FILE CONTROL)

Trường thông tin	Nội dung chi tiết
Tên tệp tin (File Name):	variables.tf
Đường dẫn trong Repo (Path):	terraform/variables.tf
Công nghệ:	HashiCorp Terraform (Infrastructure as Code - IaC)
Mô tả chức năng:	Khai báo các biến cấu hình tham số hóa hạ tầng (Ports, Credentials, Environment)

1. Toàn văn mã nguồn kịch bản `terraform/variables.tf`

```
# =====
# Terraform Variables - HCMUS-LDMS Infrastructure as Code (IaaS)
# =====

variable "app_env" {
  description = "Target application environment (development or production)"
  type        = string
  default     = "production"
}

variable "project_name" {
  description = "Project name identifier used for container and network naming"
  type        = string
  default     = "ldms"
}

# --- Database Variables ---

variable "postgres_user" {
  description = "PostgreSQL superuser username"
  type        = string
  default     = "ldms"
}

variable "postgres_password" {
  description = "PostgreSQL superuser password"
  type        = string
  default     = "ldms"
  sensitive   = true
}

variable "postgres_db" {
  description = "PostgreSQL default database name"
  type        = string
  default     = "ldms"
}

variable "postgres_port" {
  description = "Host port exposed for PostgreSQL"
  type        = number
  default     = 5434
}

# --- Object Storage Variables ---

variable "minio_root_user" {
  description = "MinIO root administrator username"
  type        = string
  default     = "ldms"
}

variable "minio_root_password" {
  description = "MinIO root administrator password"
  type        = string
}
```

```

    default      = "ldms12345"
    sensitive    = true
  }

  variable "minio_api_port" {
    description = "Host port exposed for MinIO S3 API"
    type        = number
    default     = 9002
  }

  variable "minio_console_port" {
    description = "Host port exposed for MinIO Web Console"
    type        = number
    default     = 9003
  }

  # --- Service Ports ---

  variable "api_port" {
    description = "Host port exposed for FastAPI Backend API"
    type        = number
    default     = 8000
  }

  variable "web_port" {
    description = "Host port exposed for Nginx Frontend (HTTP)"
    type        = number
    default     = 8080
  }

  variable "web_ssl_port" {
    description = "Host port exposed for Nginx Frontend (HTTPS)"
    type        = number
    default     = 8443
  }

  variable "grafana_port" {
    description = "Host port exposed for Grafana Dashboard"
    type        = number
    default     = 3000
  }

  variable "prometheus_port" {
    description = "Host port exposed for Prometheus Metrics"
    type        = number
    default     = 9090
  }

  variable "mailhog_ui_port" {
    description = "Host port exposed for MailHog Web UI"
    type        = number
    default     = 8025
  }

  variable "live_app_url" {
    description = "Public live deployment URL"
    type        = string
  }

```

```
default    = "https://hcmus-projectmanage-lab.vercel.app"  
}
```

BẢN IN KỊCH BẢN HẠ TẦNG TERRAFORM IAAC:

outputs.tf

Hệ thống Quản lý và Số hóa Tài liệu Thư viện HCMUS (HCMUS-LDMS)

THÔNG TIN TẬP NGUỒN (FILE CONTROL)

Trường thông tin	Nội dung chi tiết
Tên tệp tin (File Name):	outputs.tf
Đường dẫn trong Repo (Path):	terraform/outputs.tf
Công nghệ:	HashiCorp Terraform (Infrastructure as Code - IaC)
Mô tả chức năng:	Định nghĩa các đầu ra thông tin hạ tầng sau khi triển khai (Live URLs, API Docs, Monitoring Endpoints)

1. Toàn văn mã nguồn kịch bản `terraform/outputs.tf`

```
# =====
# Terraform Outputs - HCMUS-LDMS Infrastructure Endpoints
# =====

output "live_deployment_url" {
  description = "Public live production application URL"
  value       = var.live_app_url
}

output "local_web_url" {
  description = "Local Nginx Web Frontend URL"
  value       = "http://localhost:${var.web_port}"
}

output "local_api_url" {
  description = "Local FastAPI Backend URL"
  value       = "http://localhost:${var.api_port}"
}

output "local_api_docs_url" {
  description = "Local FastAPI OpenAPI / Swagger Docs URL"
  value       = "http://localhost:${var.api_port}/docs"
}

output "minio_console_url" {
  description = "MinIO Object Storage Web Console URL"
  value       = "http://localhost:${var.minio_console_port}"
}

output "mailhog_ui_url" {
  description = "MailHog SMTP Web UI URL"
  value       = "http://localhost:${var.mailhog_ui_port}"
}

output "grafana_dashboard_url" {
  description = "Grafana Monitoring Dashboard URL"
  value       = "http://localhost:${var.grafana_port}"
}

output "prometheus_metrics_url" {
  description = "Prometheus Metrics Server URL"
  value       = "http://localhost:${var.prometheus_port}"
}

output "postgres_connection_string" {
  description = "PostgreSQL Local Database Connection String"
  value       = "postgresql://${var.postgres_user}:${var.postgres_password}@localhost:${var.postg
  sensitive   = true
}
```

BẢN IN KỊCH BẢN HẠ TẦNG TERRAFORM IAAC:

terraform.tfvars.example

Hệ thống Quản lý và Số hóa Tài liệu Thư viện HCMUS (HCMUS-LDMS)

THÔNG TIN TẬP NGUỒN (FILE CONTROL)

Trường thông tin	Nội dung chi tiết
Tên tệp tin (File Name):	terraform.tfvars.example
Đường dẫn trong Repo (Path):	terraform/terraform.tfvars.example
Công nghệ:	HashiCorp Terraform (Infrastructure as Code - IaC)
Mô tả chức năng:	Tệp cấu hình biến mẫu cho môi trường Production / Staging

1. Toàn văn mã nguồn kịch bản

terraform/terraform.tfvars.example

```
# =====  
# Sample Terraform Variables - HCMUS-LDMS Infrastructure Customization  
# =====  
  
app_env           = "production"  
project_name      = "ldms"  
postgres_user     = "ldms"  
postgres_password = "ldms"  
postgres_db       = "ldms"  
postgres_port     = 5434  
  
minio_root_user   = "ldms"  
minio_root_password = "ldms12345"  
minio_api_port    = 9002  
minio_console_port = 9003  
  
web_port          = 8080  
web_ssl_port      = 8443  
api_port          = 8000  
grafana_port      = 3000  
prometheus_port   = 9090  
mailhog_ui_port   = 8025  
  
live_app_url      = "https://hcmus-projectmanage-lab.vercel.app"
```

← CD

✔ Merge pull request #54 from se-bros/feat/iaac-terraform-evidence #2

Re-run all jobs ...

Summary

All jobs

✔ Build & Verify Artifacts

✔ Verify Infrastructure (IaaS)

✔ Deploy Application

✔ Send CD Email Notification

Run details

Usage

Workflow file

Triggered via push 1 minute ago

NguyenAn0808 pushed a173a03 main

Status

Success

Total duration

40s

Artifacts

—

cd.yml

on: push

✔ Build & Verify Artifacts 17s

✔ Verify Infrastructure (IaaS) 9s

✔ Deploy Application 6s
<https://hcmus-projectmanage-lab.vercel.a...>

✔ Send CD Email Notification 10s

Annotations

4 warnings

⚠ Verify Infrastructure (IaaS)

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, hashicorp/setup-terraform@v3. For more inf...
[Show more](#)

⚠ Build & Verify Artifacts

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-node@v4. For more informatio...
[Show more](#)

⚠ Deploy Application

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4. For more information see:

[se-bros/hcmus-projectmanage--lab]  CD (production) - main: Merge pull request #54 from se-bros/feat/iaac-terraform-evidence  Summarize this email



tuananh835.nta@11103689.brevo.com

To: tuananh835.nta@gmail.com;  AN TIẾN NGUYỄN AN; anhkhoea515.dev@gmail.com; +2 others

Sat 8/22/2026 2:06 PM

 Some content in this message has been blocked because the sender isn't in your Safe senders list.

Trust sender

Show blocked content

se-bros/hcmus-projectmanage--lab vừa triển khai lên môi trường **PRODUCTION**.

 **Live URL:** <https://hcmus-projectmanage-lab.vercel.app>

- Branch / Tag: main
- Environment: production
- Commit: [a173a03](#)
- Author: Nguyen An
- Message: Merge pull request #54 from se-bros/feat/iaac-terraform-evidence Feat/iaac terraform evidence
- Workflow run: <https://github.com/se-bros/hcmus-projectmanage--lab/actions/runs/32558692105>
- Job run:
 - build-verify:  success
 - infra-verify:  success
 - deploy:  success
- Status:  success

 Reply

 Reply all

 Forward

 `ci: integrate terraform iaac validation into ci and cd workflows` ✓ 06124fd

 vercel Bot deployed to **Preview** 3 minutes ago View deployment

 Unsubscribe

You're receiving notifications because you authored the thread.

1 participant



 Lock conversation

 This branch was successfully deployed Show environments

1 active deployment



 All checks have passed 1 skipped, 5 successful checks

1 skipped check

  CI / notify (pull_request) Skipped 3 minutes ago

5 successful checks

  CI / backend (pull_request) Successful in 24s

  CI / frontend (pull_request) Successful in 34s

  CI / Terraform IaC (pull_request) Successful in 7s

  Vercel — Deployment has completed

  Vercel Preview Comments —  No unresolved feedback

 No conflicts with base branch
Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

se-bros / hcmus-projectmanage--lab

Q Type / to search

<> Code

Issues

Pull requests 3

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

← CI

Feat/iaac terraform evidence #66

Re-run all jobs

Summary

All jobs

backend

frontend

Terraform laaC

notify

Run details

Usage

Workflow file

Triggered via pull request 2 minutes ago

NguyenAn0808 synchronize #54 feat/iaac-terraform-evidence

Status

Success

Total duration

38s

Artifacts

-

ci.yml

on: pull_request

backend24s

frontend34s

Terraform laaC7s

notify0s

frontend summary

Vitest Test Report

Summary

Test Files: 18 passes · 18 total

Test Results: 80 passes · 80 total

main.tf

terraform > main.tf > ...

You, 49 minutes ago | 1 author (You)

1 # =====

2 # Terraform Main Configuration - HCMUS-LDMS Infrastructure as Code (IaC)

3 # =====

4

You, 49 minutes ago | 1 author (You)

5 terraform {

6 | required_version = ">= 1.5.0"

You, 49 minutes ago | 1 author (You)

7 | required_providers {

You, 49 minutes ago | 1 author (You)

8 | docker = {

9 | | source = "kreuzwerker/docker"

10 | | version = "~> 3.0"

11 | }

12 }

13 }

14

15 provider "docker" {}

16

17 # --- Isolated Bridge Network ---

18

You, 49 minutes ago | 1 author (You)

19 resource "docker_network" "ldms_network" {

20 | name = "\${var.project_name}_network"

21 | driver = "bridge"

22 }

23

24 # --- Persistent Storage Volumes ---

25

You, 49 minutes ago | 1 author (You)

26 resource "docker_volume" "postgres_data" {

27 | name = "\${var.project_name}_postgres_data"

28 }

29

You, 49 minutes ago | 1 author (You)

30 resource "docker_volume" "minio_data" {

31 | name = "\${var.project_name}_minio_data"

32 }

33

main* 01:21

Launchpad

34 0

Select Postgres Server

You, 49 minutes ago

NguyenAn0808 (49 minutes ago)

Ln 14, Col 1

Spaces: 2

UTF-8

LF

Terraform

Antigravity - Settings

Prettier

main.tf main-tf.png U outputs.tf X

```
terraform >  outputs.tf > ...
```

You, 50 minutes ago | 1 author (You)

```
1 # =====
2 # Terraform Outputs - HCMUS-LDMS Infrastructure Endpoints
3 # =====
4
```

You, 50 minutes ago | 1 author (You)

```
5 output "live_deployment_url" {
6   description = "Public live production application URL"
7   value       = var.live_app_url
8 }
```

You, 50 minutes ago | 1 author (You)

```
10 output "local_web_url" {
11     description = "Local Nginx Web Frontend URL"
12     value       = "http://localhost:${var.web_port}"
13 }
```

You, 50 minutes ago | 1 author (You)

```
15 output "local_api_url" {
16     description = "Local FastAPI Backend URL"
17     value       = "http://localhost:${var.api_port}"
18 }
```

You, 50 minutes ago | 1 author (You)

```
20 output "local_api_docs_url" {
21     description = "Local FastAPI OpenAPI / Swagger Docs URL"
22     value       = "http://localhost:${var.api_port}/docs"
23 }
```

You, 50 minutes ago | 1 author (You)

```
25 output "minio_console_url" {
26     description = "MinIO Object Storage Web Console URL"
27     value       = "http://localhost:${var.minio_console_port}"
28 }
```

You, 50 minutes ago | 1 author (You)

```
30 output "mailhog_ui_url" {
31     description = "MailHog SMTP Web UI URL"
32     value       = "http://localhost:${var.mailhog_ui_port}"
33 }
```

main* 01 21 Launchpad 34 0 Select Postgres Server

👤 You, 50 minutes ago 👤 NguyenAn0808 (50 minutes ago) Ln 9, Col 1 Spaces: 2 UTF-8 LF {🔗} Terraform Antigravity - Settings ⌵ Prettier 🔔

```
PS D:\Project\hcmus-projectmanage--lab> cd .\terraform\
```

- PS D:\Project\hcmus-projectmanage--lab\terraform> .\terraform.exe init

Initializing the backend...

Initializing provider plugins...

- Reusing previous version of kreuzwerker/docker from the dependency lock file
- Using previously-installed kreuzwerker/docker v3.9.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

- PS D:\Project\hcmus-projectmanage--lab\terraform>

```
# docker_volume.grafana_data will be created
+ resource "docker_volume" "grafana_data" {
  + driver      = (known after apply)
  + id          = (known after apply)
  + mountpoint  = (known after apply)
  + name        = "ldms_grafana_data"
}

# docker_volume.minio_data will be created
+ resource "docker_volume" "minio_data" {
  + driver      = (known after apply)
  + id          = (known after apply)
  + mountpoint  = (known after apply)
  + name        = "ldms_minio_data"
}

# docker_volume.postgres_data will be created
+ resource "docker_volume" "postgres_data" {
  + driver      = (known after apply)
  + id          = (known after apply)
  + mountpoint  = (known after apply)
  + name        = "ldms_postgres_data"
}
```

Plan: 14 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ grafana_dashboard_url    = "http://localhost:3000"
+ live_deployment_url     = "https://hcmus-projectmanage-lab.vercel.app"
+ local_api_docs_url      = "http://localhost:8000/docs"
+ local_api_url           = "http://localhost:8000"
+ local_web_url           = "http://localhost:8080"
+ mailhog_ui_url          = "http://localhost:8025"
+ minio_console_url       = "http://localhost:9003"
+ postgres_connection_string = (sensitive value)
+ prometheus_metrics_url  = "http://localhost:9090"
```